ActInf GuestStream 056.1 ~ Grégoire Sergeant, "Agency with structured latent state-spaces"

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foreign it's September 15 2023 and we're at the Active inference Institute in active guest stream number 56.1 today we have Gregory Sergeant Petri and we'll be hearing a talk followed by a discussion on Geometry of world model influences behaviors first there'll be a presentation and then any comments that people have in the live chat or any other questions it'll be great to talk so thank you a lot for coming really looking forward to this so peace to you May thank you very much Daniel for the invitation first it's really I'm very happy to to be able to present this this work here so it's a it's a work around the like some models uh of Consciousness at least some some computational models of Consciousness some part of Consciousness and how it can help to generate a different kind of behaviors especially when we let's think about let's say artificial agents so it's a um it's a work in collaboration with uh so David Davidoff Kenneth Williams in fact it's a it's not just one work it's a collection of of works that been have been on for like uh already around 10 years so if you want to know more about like this group and the work that we're doing you can go on the page on my personal webpage and there's a link to PCM dot HTML and there's a summary of all the work we've been doing and some uh summary of the work we've been doing and also of some articles that can be relevant on this subject so today we'll focus more particularly on two uh two articles so one that is accepted and will appear very soon the other one need to resubmit which uh give formulation of let's say autonomous agents or like let's say mathematical formulation of agents that have a world model that is structured geometrically in such a way that this world model captures some features of Consciousness so the first article is let's say the that we we view all the experimental results that we have and the like most recent formalism that we have on our work which is to the literature on pomd partially terrible monitoring process which is optimal control stochastic optimal control and we just try to tweak a bit and this formalism to uh introduce the ideas with how you can include inside of the world model of the agent some ideas on Consciousness and still continue to have algorithms to do inference to find Optimal policies and everything and the second article focuses more particularly on how uh changing the world model of the agents can change its behavior in particularly the relationship the geometry of the world model the way that it perceives Etc and the way it acts with respect to uh for agent strategy so the way that it looks for something so it will change its behavior and looking for something so I will start by presenting a bit what uh what are all these terms or World model what you needed what it is what is a foraging strategy what it is to how you how can you define an agent that is looking for a certain object in particular what is an exploration based on curiosity or say more technically epistemic value so let's consider the setting where you have a let's say a real world so it's a space that surround us through each space you do a setup where you have an agent which is let's say a solid it has a solid frame so it has a center of three axis but it's in front but it's on its side where it's above above it and it's looking for an object o which is itself a solid inside of R3 and all the configuration of the agents and the objects they are defined by let's say a reference frame that is a external and that the that characterizes completely their configuration inside of this world now the agent a what it wants to do is to find for the object o but it doesn't it can only have like some noisy observation of where the object o is so to be able to find all it needs to have some a priorian world o should be to be able to plan the consequences of its actions with respect to where or will be once it has moved and to update and to be able to plan what the how the observation can act with respect to updating its prior so in other words so you have have an agent and the agent is looking for oh and things that always for example in this direction on the right so it's going to move towards the right and makes an observation if it doesn't CEO it's going to change its belief on where o is so this is a very standard formalism that you can find like in pomdp stochastic optimal control but the one other point we want to to to Really emphasize is the fact that the agent a has its own frame of reference in the way that it's going to give the coordinates coordinates of the object the way that it will Trace worthy object is is using its own way of computing the the coordinates of O so it has its own way to measure where o is so it doesn't have to refer to a global frame a global coordinate system to be able to know where o is so for example I am always centered on myself when I move I see the object I I don't think about me moving but I think about the object moving and this is because when I move I change my frame so the way I'm going to reference the object is going to change accordingly and the point we want to introduce is the fact that you can have several ways of defining for the agent to Define its own coordinate system with respect to its environment and this is a key feature of that we want to capture like in terms of of modeling but that's also wishing that it is something that is very structuring with respect to models of Consciousness because when you consider like more generally an agent that that had that is let's say that has a world model uh on in which it has its own let's say uh it has its own point of view perspective on this world model you kind of have to start tackling the the problem of how to introduce a point of view on the environment of the agent how can you introduce in the way that it's uh it is going to encode its environment a certain point of view and in particular if you think like about it in terms of the role of space to be able to take into account your own perspective on the environment you can see that if you take away this like everything that fills up the space you keep just the space itself it's it's uh the space allows you to uh let's say to structure the way that you are going to fill in some informations about the environment and it's centered on your own point of view in a way that when you're going to act you're still going to keep this still says this the the like the unfilled space the St you you're still going to keep the stay same way of modeling this space the space will stay the same when you move when I move the space is the same but I talk into account the fact that I can move from one point to another in this space without changing the space and something that is also very important is that with respect to any movement that I can do the space doesn't change so there's no particular point in space with respect to the movements that I do another way another property that is very similar is that instead of considering just my actions and the way that I can change my frame while I'm doing an action you can also Imagine taking the frame of somebody else so it means that you can you see that the space that you you use to be able to structure your environment doesn't change when you take the perspective of somebody else for example if you imagine yourself as a solid agent changing by your rotation your own like coordinates your own solid frame and translating okay so this is uh something that we will use in fact as a definition for the state space the the way that the agent encodes its environment and we will see that there's a natural notion to be able to that encodes this idea that you can change perspective on your environment on the world model that you have of your environment simply by introducing something that's very well known which is the notion of a group acting under a certain space so this is just to sum up a bit the what we're trying to do so we're just trying trying to do like exploration inside of an environment but we're adding an additional information which is that the way the agents is going to encode its environment takes into consideration the fact that it can change its reference frame on this space without singling out any point okay so what do you mean like perspective taking in like more precisely so how do we go from The Real World so the let's say the world and code the world described by the external like a frame that allows you to say the agent is here and the object is here from the internal world to of the agents so you just Define the map that will take into consideration that the agent can have a first perspective on this environment the simplest case in the occasion case is just rewriting the coordinates a bow inside of the frame of the agent reference rate of the agent and then once the agent moves its frame solid frame in the real world changes and this induces indeed the transformation inside of its internal World which will be in a fine transformation okay but you could imagine also having other Transformations I don't know if you can see my my mouse probably can you see my mouse or not no okay so okay so the map C PSI can also be something else than a defined transformation it could be like any kind of transformation for example in a fine case the apparent volume of the object doesn't change but if you change this map and consider for example the projective transformation as we will describe later you can add that the size apparent size of the object changes so we we went from something that is a bit like already like very well known and not not very Innovative which is that you can rewrite like motions depend like the emotions depends on the way it depends on the frame that you choose to describe them into seeing that you can in fact change some frames to be able to account for some uh like perceptive property and what we asked also for the agents that we consider is that they can imagine the consequences of their moves which on their uh future observations so there's something just to say that the agents that we consider have a notion of agency which is that they can plan the consequences of their actions so that they can choose the best action with respect to a certain reward in particular the reward that we consider is the trying to maximize a certain surprise so to sum it up we have an agent that is looking for certain objects so it has belief of where the subject is and the beliefs lives on on an internal State space so the world model of the environment it can make observation which allows it to update uh its beliefs and then it can predict the consequences of its actions so that it can plan the way that it should act with respect to a certain objective so formally this is simply the notion of markup decision process or more generally like notion of partially observable markup decision process because we do not have a complete information of the environment but also but only some observations that are that are limited and uh it's very related and I mean there's a strong Duality between Po mdp and active inference so there's really a strong link between both so now what I did in the way that I presented it so I presented it with the idea I tried to present the idea of what we're doing so the general context pomdps agents that are planning uh their action with respect to certain objective here the objective is to find a certain object and uh I give uh the formal setting for defining it which are ndpmvpn now I will go one step further and Define it explicitly and I will continue like this approach for the second part where I will also present our results and our specificity where else developed with the general statement then a bit more precise and then really digging digging into the result so here the classical way to define the markup decision process is to consider that you have a set of configurations of the environment which is the state space of the world model in the way and at least the way that you encode your environment so this is something on which you can act because the agent when it makes an action it changes the state of the and the actions the way that the actions change the environment is with respect to a certain uh Markov kernel probability kernel so it's stochastic which means that each actions changes the state of the environment but you don't know completely I guess not deterministic it you allow some errors in the way that it acts on the and you have a reward form a function that is associated to the actions that you can do at 20 and also the state of the the agent that time t and partial observable partially observable markup decision process is the same thing that a markup decision process but you authorize to say that your observation on your environment are only partial which means that you do not have a pre-say precise information on the states of the environment so the way that you need to relate observation and states is through also a probability kernel which tells you that uh if you're at state for example s you would expect to make a certain observation let's say of if you if you think the object is at The X you expect to see the object attacks but with a certain error because you know that your sensors are also random they have some they have they can make some mistakes and you keep similarly a reward function so it's the same thing that an mdp Markov decision process but usually also allow to have some to get information on your environment through observations that are not complete in fact there's a formulation in terms of partially observable markup decision processes our particular case of markup decision processes which is called belief Markov and belief and DPS Billy markov's decision process and the only difference that you have between Minecraft process and belief mdp is the fact that the state space is necessarily continuous in the second case so po ndps can be seen as a particular place of ndps we said that the actions of a pomdp act on the state space and you have you account for the consequences of the actions only through observations so graphically it means that you have the state space of your so the world model your environment X on which you you like you reference for example where the object is so the position of the object and then when you act it induces a consequence on the position of the objects at time t plus one and then you can make an observation of who you think the object will be or you can make an observation with respect to where the object is depending if you're planning into the future or if you're really doing the action uh you're you're implementing the consequence of the actions in the real world the thing that we're trying to do in our setting is to replace simply the actions by a change of perspective like I told you like when the agent asks it's coordinate system on the environment changes so you can always see it as in a way passively where you just see it as a change of coordinates so we want to say that instead of considering actions on the environment we have a space on which there's a natural notion of changing of reference frame and actions are simply certain kind of changes of frames we also allowed to have some actions that are do not correspond to changes of frames we just say that we include the possibility that some actions are changes of frames and the changes or frames are have something that is related to all the Transformations that are internal at the for the agents so for example things that it knows that a priori it knows it's encoded in the way the agents will interact with the the way that we do it a bit more formally is that we say that changing perspective is simply through the action of a group so for example in the affine case when the the agents moves it changes a frame is a defined transformation and we say that the world model so the state space is simply a space on which the group acts so formally what does it mean that the group acts on the space is just saying that you for you have space s and Group G there's an application that goes that takes an element from G and S and that sends back s in other words for every GI you can associate a function from S2s and you assume well so that it has good properties which is that it satisfies equation one and that uh if you don't move you stay at the same place so this is like just the notion of a g space a space a group acting on the space but now if we want to include this in ndps and P1 DPS we just assume that the state space is a g space that some actions are some elements of the group and that when we choose the actions that corresponds to the element of the group that corresponds to the way that the group acts on the element so it's very like there's nothing like very convoluted it's just saying that you define a collection of functions which you see as changes of frames and the way that they act so the probability kernel from the state space to the States states that time T2 time t plus one after reaction uh after the change of perspective is simply through the way that the function changes the state space so it's just kind of a way of reformulation but what is really hidden behind here is that we have the structure of the group and that we don't consider any kind of actions we assume that there is more structure in the actions that we can consider and then it's defined that is that it's uh in like in the way that is uh it's that it's encoded inside of the geometry of the space that we have so we don't separate any more action and state space we say that the state space in its geometry encodes already certain kind of actions which are the change of perspectives so there are two cases that we consider the first case is where the status is The Arcadian space and gdfn transformation it corresponds to like I find Transformations translations and rotations of the agents and changing rewriting the coordinates of the objects inside of the ref solid reference frame of the agent and the second case where uh the first the state spaces the projected space and the group is a projected transformation we described how what are the generative models that we consider and how we can include so the inside of the classical theory of pomdps the fact that you can take a perspective inside on your world uh on your on your environment now we will introduce uh like a classical notion of epistemic value which will allow us to Define what is uh Behavior what is an exploratory Behavior with respect to curiosity so how what it is foreign agent 2 uh exploits an environment based on curiosity curiosity or let's say the drive for exploration is uh uh quantity that will it's a it's a quantity that you will um Okay so it's uh okay it's so how do you define it just with like very generally so you start with the agent has a prior on the state of its environments then it plans the consequence of one of its move step so this changes the prior that it has on the environment because there's action on the environment so we get a new prior and now it's going to make an observation once so it imagines that it's going to make an observation once it makes an observation it's there's an aposteri the priorities of data so you get an Apple Theory the way that you Define curiosity or let's say epistemic value is how informative is the observation that you will do how far is the apostillary from the apiary but this you cannot do it for one given observation because the way that you compute It Is by planning what will happen at the next step so you need to consider all the possible observation that you will make so the observations are themselves stochastic with respect to the way that you consider which is actually a priori on the state space of a Time t as we said for us actions are changing the frame so we can also Define epistemic value four frames for changes of frames and in particular something that we gain that is I think interesting is that now we have a function that is defined on the on the group that can be a continuous group so you can allows you have for example if you want to maximize it you you can or minimize it and it depends how you see it you can watch you you couldn't do gradient descent for example so you can have more analytical tools because you're on the space that is continuous and has some structure so now to give the formal definition of epistemic value so as I said it's based on the quantity c c that I will Define now so if you have yourself if you give yourself a prior on a space X and you give yourself a probability kernel so stochastic map from X to Y stochastic map is simply saying that for any X you will associate the measure on y then you can get a joint Distribution on X and Y which is simply given by the product p y doing x times the prior in fact the quantity C is simply the mutual information between X and Y which is saying how far is the joint distribution with respect to the uh the independent distribution the product of the marginal distribution next and on y but this is like you you see it it appears everywhere the mutual information so this formulation is based on the paper that is uh from a fist on the null active inference and a pessimic value the mutual information is something that appears everywhere but I think that there's a very nice re-expression of the mutual information which allows to give a better interpretation of this quantity and which is the interpretation I was discussing in the previous slide so if you rewrite Mutual information you can always see it as the coolback library distance between the posteriorly for a given observation and the prior pullback distance being a way of computing how far you are how far the two distribution are but you need to look at the expectation with respect to the observation that you make so it's exactly what I was discussing before so then uh uh this is like for any kernel any kernel from X from y x to Y with a prior on x but as we were discussing here you have to take into con so this would be the kernel but you have to take into consideration that you can do actions and let your prior r on X so the way to the compute epistemic value in for this kernel here when having only your the prior on X is that you propagate the prior buy the action on X1 then you get the prior on X1 and you can Define the epistemic value for this prior and Markov kernel which responds to like the randomness of your sensors which is always fixed and this is very important this one doesn't change even if you change frames the kernel that you have relating your prior what you think about the state space so relating the state space and the observation never changes and so especially this means that we after a certain action or after a certain change of frame and get a joint Distribution on X and Y which is the following one here so this corresponds to the prior propagated on X1 and then the epistemic value is simply the neutral information of this joint distribution so now how does the algorithm work the algorithm that corresponds to uh defining an exploratory behavior for an agent that is looking for so he said it started with a prior of where o should be then you maximize curiosity based on some changes of frames that are around the identity Elements which correspond to not changing frame then you get an action that you can apply you propagate the prior to the next step with this action and then you just update your priori with respect to a certain observation and then it Loops back so this is the algorithm that we consider uh in the second paper that I that was listed in the in the presentation so which corresponds to having an agent that is looking for a certain object is it's um behavior is defined by an exploratory is is driven by exploration uh driven by curiosity taking into account that its state space is structured by the action of a group so that the state space is structured by the fact that it it corresponds to all the possible ways of like changing frames so it has inside of the state space inside of the geometry of the state space all the possible ways of changing frames and uh this is this is like explicitly what we do so I mean in this context this is the algorithm that we consider and nothing else then we get a very interesting result at least I find I mean it's not it's interesting and like in the way that it's presented here I find it interesting and then I mean if you go more into detail and it's it's because the occlusion case is very particular but what you get is that if you look at the behavior of an agent that is driven by exploration by curiosity uh but that has a state space structured by occlusion transformation or clearing transformation so the first case where the agent is simply uh encoding its environment in its reference frame but nothing else then it doesn't need to move in the second case where the changes of charts are given by projective Transformations so the way it encodes to it's in transformation they take to account a perspective a projective uh deformation of its environment then it will always try to get closer to the object so you have two very separate behaviors so this is something that is I think it's a very interesting to to note so before going into the details of uh how you can improve the statement I will say a bit more about why it's more it's an interesting perspective point of view on the on the subject because and how it goes a bit further than simply considering this very simple setting what you could imagine is that encoding the fact that the agent encoding the actions of the agent directly inside of the state space of the Asian through geometry could be a way to stabilize the representations of the agent and this is something that we're working on now and there's already already literature in this direction okay so now let us proof uh the statements and to prove the statement we need to give a formal like a precise uh statement so what more particularly what we were able to show is that if you assume that the agent has us move staying still so it's allowed to stay still then if uh it's it's uh changes of frame are given by actually by a fine the agent stays still now in the projective case if you assume that the agent is always looking in the direction of the object it will always try to get closer to the okay so the idea of the proof is that in the what what place the the role for discriminate for for the tribe of the agent is how big the agent is appearing to in the reference how is the size of the object in the reference frame of the agent and it's how big it is it appears to be to the agent so in the first case the volume of the object in the internal space of the agent doesn't change so it doesn't need to move in the second case if it makes a move what is informative is to try to make the object bigger because once it's bigger the apostillary will be further from the prior with respect to a certain observation so it will always privilege moves that allow to make the object look bigger and in this case what you can do is uh you can you can show that this corresponds in fact two actions or change of frames that gets you closer to the object okay so how to improve the results so as we said we consider a change of frame from The Real World to the internal World here I didn't consider changes of perspectives with respect to moves here is simply how you relate the real world to the internal world in the euclidean case it's clear it's clear it's just the way that you write the coordinates of the object in the solid frame in the projected case is clearly not it's not obvious because there are several ways that you can relate the solid frame of the agent to the projective frame so one way that we decided to do it is we give some set of actions that we consider to Media let's say uh coherent with our own experience of space which is that we feel that we're always centered on ourselves that the access of the site frame of the agents inside of the like 3D space so what is in front of it or what is on the right where the stops are preserved that there's no point in front of the object that appears to be at Infinity and that that near to the center of the object the volumes are preserves so when you do this so this uh this formulation like this result is in in this article here so the way that we relate Some solid frames like some solid referencement of the agent to some protective Transformations are in this article in particularly in proposition A1 where we show that this set of actions limits the set of projective transformation we can consider and we will only have this and so the change of frames from the external world to the internal world is given by rewriting the coordinates of the agent of the object in the solid stream of the agent and then applying this projective transformation so now uh what we defined here is are the maps that relate the observation of the agents Zach so the we we defined the maps in the occlusion and projective case that relate the observation of the agent the observation of the object to the way that it represents this this object inside of its internal world here this is simply like the pomdp that we defined before and now once the agent doesn't move it changes its equation reference frame so there is a solid reference frame so that from going from one side reference frame to another side reference frame you can by applying the projective transformation we had before Define another one that we call PSI that goes from the state space at Time Zero to the state space after action after moving very what is very important is that we consider uh the Markov kernel Associated to a noisy observation to be of this shape here which is that if you know about the object if you think the object is at point x then you believe that the observation will be around X in the ball uh around X for a ball of a certain radius which is a small ranges Epsilon okay and I need to charge my computer so sorry so now press uh epic value as defined it has a very simple expression so the epistemic value of the prior propagated after the transformation Phi after this changes of frame five so you have a problem here you propagate it on X1 through Phi and you compute the epistemic value for this uh for the joint distribution over X1 and Y is given by this formula here well it's simply integrating over all the possible places where the object could be times the probabilistic volume of the ball of size that sit on uh for the propagated measure and the logarithmic of the sequentity okay so this is very direct to write you can just compute it and you find this expression but what you have already here is that if you consider an oxygen transformation then the quantity Cube sine minus one B Epsilon y doesn't change so then you have the sigmatic value that is a that is constants so basically if you try to maximize the systemic value maximize this quantity you can do anything you have always one move which is not which is not to move so you don't move it's okay it's perfectly fine now in the second case we have the projective where you consider that the way you relate environment and the internal world is through a projective transformation then it's more complicated and so you need to uh use a trick which is that you know that once an observation has been made the support of the prior will be smaller and so after one step if you suppose that your Epsilon so the size of let's say noisiness of your sensor is small enough you have a support that is uh of the distribution that sufficient is small enough so that you can do an asymptotic development of the quantity in the integral so this is an equality but this is an approximation here and now this is very useful because now that you know we're like doing this in a way uh you can say that you just need to develop this quantity at the point where the object is really so if you do it you get this expression and what appears to play a role is only the determinant of uh the Jacobian or the projective transformation which here in the case will be one can be many things so here also is not very clear because how do you define CM so cm is defined as a composition of several Maps I didn't go into the details but it's corresponds to the map that are given by the changes of frame so if you have the oxygen space and the internal spaces after moving you have a change of frame in the equation space but it corresponds to also a change of frame in the internal space so if you the way you define PSI is simply saying that you inverse the projective transformation from internal world to external world to go at Time Zero then you apply the change of frames and then you apply the projective transformation to go from external to internal this formula that you have here and which is very nice with this and then you have the first term that doesn't depend on the actions that you do so you don't need to take it into account this one will be one and then you just need to compute this one in fact this one has a very simple expression so it's expression 20 23 and then here you can directly see which moves correspond to increasing this quantity and then decreasing this quantity because we want to increase epistemic value and so this this is the result this allows to prove the result that we we said in the first day like in the several slides before that in the projective case if you have enough movements if you look at the object then you're always going to go closer to the the way to interpret it is as if the agent it was a bit paranoiac or paranoid if it's like I'm very uncertain on its own beliefs so it knows the object might be over there but it's always uncertain so it needs to go check and once I check it's more certain but still as it can always be even more certain it will always try to get more and more certainty and so I only presented like in this uh in this presentation all the aspects which are more computational and what the algorithm and some like analysis of the algorithm we considered but we also have like some experiments on how this setting can allowed to generate different behaviors behaviors that we would expect explain for example some Illusions like the moon Illusions and then invite you if you're interested on this to listen on to the like online talk moc-4 conference that was in Oxford last week or to check up check out one of these three papers and I would like to thank you very much for your attention all right awesome wow very interesting and different ways than how I've seen the pomdp and related uh works so okay let's just start off with little context and then I'll read some questions and read some questions from live chat so what what brought you to study this question this way uh what question the question of uh okay did you come from a math side and find Consciousness and and geometries to be interesting or vice versa what kind of brought you to want to make this contribution so at the beginning so when I did my my uh PhD in maths I was more interested I was interested in let's say uh like all this idea of critical critical brain hypothesis so uh the way that the trying to understand the way that the brain processes information and makes it something that is uh like exploited and so they one of so the critical brain hypothesis tells you that the activity of the neurons is basically close to certain criticality criticality in the source of statistical physics if because you can model the activation of the neurons as let's say uh statistical system like an icing model but the so I worked a lot on this and then I there's another hypothesis that is uh like very common which is the Bison brain hypothesis and so the Bayesian brain hypothesis led me to active influence to learning more about optimal control by Asian fashion perspective on optimal control and uh there was uh so uh my advisor was a PhD advisor was working with David and they're still together on uh trying to uh Implement uh some aspects of consciousness uh and how it can influence influence especially with their article in the Moon Illusion so uh I was interested in knowing how uh how uh this uh kind of let's say uh ideas uh simply like in a very naive way they interact with the Bison brain hypothesis and then I continued in fact like one of the lines of my research is structured like algebraically structured statistics or machine learning and so the way that they see it like very geometrically in fact or let's say geometrically or algebraic it's not the same thing but it's very related it's something that I I wanted to uh to to understand a bit better let's set you to make it in a formal setting so that it's simply a particular case of what are what is in the literature so it's specifying what exists in the literature and this took some time because I was more on the active inference like the free energy principle side so I had to read more about like Optimal control sarcastic optimal control qmdps and understand that in fact what we're doing is simply adding more structure on the latent space of of the agent and the state space of the agent and uh doing this allows to ask the question like why is this useful why having a state space that encodes different perspectives so the motivation comes from the Consciousness and study cognitive Sciences but why it can be useful for Robotics and it's always been like my motivation study uh some statistical statistical models more structured models with more our priori that can be useful uh for understanding the behavior of a closed system like an agent like a collection of neurons like even like molecular machines okay so that's a very long answer but it's okay cool so you focused on the spatial movement epistemic foraging case is there something special about space or can we also think about this perspective taking in terms of for example semantic or a narrative reference frame so the this is I think it's a very very important question because up to now all the work a reference will need to space so the fact is based in terms of the 3D space and not space in terms of geometry and the fact of writing it as geometry allows to get out from the classical point of view as space as a 3D space because you see that there's more and more like for example in the in geometric deep learning there's more and more the use of uh space to encode environments of certain objects and these objects are not necessarily uh had the three the three-dimensional structure you can have objects that have higher groups of ignorance so I think that the it's it's indeed using geometry for other contexts and it's really the aim of trying to go through this more General formulation is to be able to apply it to real world models what I mean by real world models it means the ones that are learned to be able to have an agent in an open environment that will learn the way uh it would learn its generative model but with they are priori that it can take a perspective on its environment and see what it can do so we really want to get out from the 3DS at least for me go out from 3D space and go to like implementing it completely in autonomous agents okay to kind of follow on that there's a question to chat great talk I also wonder whether it applies to any modality not just visual spatial but also text yeah so it's for other modalities like sound for example what I don't know for applying to text so there's a lot of work now on uh large language models and like of this idea of prompting and being having these models being able to have some kind of imagination so you would like to have this kind of these ideas applied in this context but for now I mean it's not the line that we're trying to that I'm trying to do so I'm trying to stay on deep learning so basically take standard data set without considering uh uh like text and just try to see how these ideas like in geometric deep learning can stabilize representations so it's not the same that is a lot of others it's clearly other modalities especially if we consider like multi-model integration like you you try to rebuild the state space and you don't want to see it only as a vector space but you wanted to see it got more structure because you want to force the way like you have constraints be you know appear only the constraints that you're going to have on the way that you can take a perspective on one volatility or the other but it's not for text at least not matter I think it maybe it could be used for text but but it's not what I'm doing now yeah okay so it's I I I I just wanted to rebound I have the lack of collaborating with real mathematics some guy with intuition that I've found the right people to to do the work uh regarding multimodality this is important to understand that this is not about vision this is about spatial recognition it's supermodal the claim is that vision is just one particular way of integrating information indeed in an obviously projective manner but that's integrated in a much larger field of experience than the field of view or the visual field and obviously proprioception touch when you build a representation just by touching something and you get the 3D representation of the scene automatically in your mind hearing to the extent that it's about Source localization and building spatial uh uh uh representation all of that stuff that's the claim of the theory is integrated in this projective space they are priors from memories there are stuff from Vision staff from addition stuff from popular reception in the reception you name it all senses contribute to it so it's not Vision the claim is that projective 3D projective geometry in that case is beyond Vision vision is just actually a slave to that Supreme other representations that's the claim um awesome all uh yeah please that's important because we haven't choose I think we like we let's say we work like a group there are several like different perspectives for me I'm more on the computational side so it's important to have that color because clearly something I won't be able to answer cool here's a nice following question from Vladimir in the chat they wrote if the EG visual sensors have variable resolution for example higher resolution in the center this can naturally lead to curiosity based change of orientation response within your framework so is it the question or an affirmation so it's for EG right so if there's a variable if there's variable sensor Precision for example higher Precision in the sensor might you see any resulting curiosity Associated change merely based upon the asymmetry or the structure of the sensor Fields okay so I shouldn't say it but I will still say it because I don't know if the people will go out one day or nothing this is more attention this is more attention than curiosity so it's more like uh so you can act on your sensors so that you can uh change the way that you integrate them and this really acts as a form of attention so I really went into it what do you think what is the interplay between attention and curiosity uh I mean it's it's the way you're going to move basically on the the way the way that it deforms the way that you move that you so the consequences of your actions are not the same so basically curiosity is what what drives actions and the way that you uh said the con so you choose your action with respect to reward which is a epistemic value and now attention or let's say changing the sensors is a way to change the consequences of your action with respect to let's say optimizing this value so it's it's like yeah I mean it's it's yeah I I don't know if I should say yes or not but okay okay let's say it's okay that's it for I think it's symmetric basically it's it acts like a sort of metric on the space on the the group directly so it means that it's going to so when you have a function you want to degrade in this engine you choose a metric and the metric is so the attention will act as a metric and so the metric will allow you to deform in fact um the the steps that you will do and so I think this is how it acts so it's very known that I mean it's known that the fact that changing the sensors is is related to attention that's not something much new but the fact that you can directly in our setting related to groups and you can relate it to uh like changing the metric on the group is something that that can be done and I think it's it's interesting um all right another question in the chat what is a formal framework for learning geometry from data how do we move from empirical data sets the files on our computers and the things that we do deep learning on take into machine learning pipelines and how do we utilize geometric approaches and formalize to the kind of analytical Precision that we saw here some of these geometric relationships so the the there's a very like uh there are several different fields of how to use geometry uh on data so there's already like I mean depends how you see geometry but one way is a TDA topological data analysis so which is basically trying to uh let's say uh do you want supervised learning by geometric Pro like stability property of of your data and try to interpret it as a certain kind of space and then you look at the whole of your space and this is something that defines your data set so this is one approach another one is that people doing many like they try to learn the so they they know that you're that the that the data lives in the low dimensional manifold and they try to learn this manifold so that's another perspective another way of doing it so there's a lot of work in this in this direction there are people who are interested in variants and equivalence so more in the geometric deep learning there are people who are interested in the same setting on appearance how to use the geometric aprioris and included in learning for a deep learning reinforcement learning and this is what we do in fact this is where one of where many people do this it's not just then we like for us we try to focus on this idea of how to export it for reinforcement learning um so in this setting epistemic value was the only driver of action is that yeah so it's kind of like a expected free energy except without pragmatic value so we only have the epistemic term remaining exactly exactly in fact if you look at in terms of optimal control and not by Asian because you know that there's this Duality that is most of the time stated in like active influence where you have basically a duality between the uh let's say uh about like a stochastic value function and the probabilistic version of value function so you can encode like on priors you can encode all your drives with priors or with uh the value function directly so rewards and so both are kind of dual and they're dual in the way and you can make like this there's Duality I mean it's I don't know if it explicitly dual like in some context this this sentence makes sense for active inference for me I'm not completely aware if the Duality is exactly formal but at least this the idea is here so there's no big difference at least the way I see it as with value function and seeing it probabilistically but the terms of epistemic value is a is a is an exploration Drive which you also find in reinforcement learning maybe not in this exact expression I think that the exact expression that was given in the paper of a 37 and all is is [Music] is on the paper it's it's um it's um it's a very uh uh like a canonical way to define a epistemic value so it's an exploratory drive you can always add to the value function you know that there's a lot of problem of trying to explore in fact uh like your environment you find the good policies like it because if you're in a state space that is continuous it's really difficult to uh to result in p1bp yes it's like if you knew which Curiosities you could get rewarded from you would have already known the answer to the search so that's that's one of the challenges with pragmatic value it converges well to expectations but then this work uh really focuses in on epistemic value and shows what it can do alone as a driver so how would you bring pragmatic value into this formalism you just added you just like it's really you can put the value functions of some of the rewards in some way and then you add and then you add the epistemic value in fact when you look at the formulation in terms of belief and DP for pumdp the Curiosity epistemic value is simply a value function nothing more it's a one-step value function but I mean this is like just playing with definitions but so you can always add a drive for uh for exploration and this is something that's really often done in uh in reinforcement learning I mean I care yes like I think it's even standard in not this way exact not exactly this expression but there's a book that is called reinforcement learning state of the art and they they introduce this exploratory Drive like this is a book maybe that has 10 years or something like this so it's something that you add up you can put you can put your the way we will do it now is like we we can we can put some drawers or with respect to preferences and then you add for example an epistemic value and you try to to just solve the optimization problem but like in terms of formalism as you look at the belief mdp it's nothing more than a certain value function so it's not so space remains when we translate through it or when we whether we're in the euclidean or in the projective setting space is basically what is not changed through action is that the case okay and so is that the weight so the so the thing is that so okay so there's a bit of technicality so the way that we use it we use a chart so we need to take for the protective plane so we need to use homogeneous coordinates so we need to take away a like a lower dimensional plane so that's why you always have the same like in the the way that we encoded we always are in R3 because we chose a chart I mean we chose a homogeneous coordinates and we use projective Transformations from one homogeneous coordinates to another homogeneous coordinate but it always lists to a Project X transformation so what is hidden in the way that we write it is in fact we don't have the same space in the first case we have a Cleveland and the second we have projective so but the way as we use charts and we like I mean we don't put all the details of the fact that for example you can compose two projective transformations in uh by even if you write it in charts in terms of homogeneous coordinates and it stays a productive transformation this is something that is okay but it's it's we just don't write it this way we stay in charts and so that's why there's a similarity between e because we want to implement it there's a similarity between the acridian projected case clear case and protective case have the same state space R3 but the space of transformation is really different and in fact they are not the same space but when it's important for us in terms of spaces just to tell yourself that if you take away everything that is inside the space all the all the like everything that populates space you take it away what what you're left with is with this kind of Concepts which already takes into account the fact that when you act you're not changing you're not changing the space and there's no single point that is singled out so this is very important because it's a very typical like it types A lot of the objects you're looking at like if you move there's no point in the space that is changing there's no point that it's single out and when you move the space it's not changing so you know that you you already have inside of your space a change of charts that is that is encoded and even more when you can imagine that you take the perspective of somebody else and this is like really something that is at the basis of the space we consider is that the space is simply a way to support the fact that you can change charts you can change perspectives and so it's including this idea inside of agency that we're trying to do so maybe I was not clear in the way that I installed it first but I think this is really opportunity it's the water we're in so it's a very interesting um way to approach it and it reminds me way back when actually almost three years ago when we discussed the projective Consciousness model and phenomenal selfhood in live stream number nine and um we talked a lot about flipping between the euclidean and the projective modes how an agent could have on one hand a space in which a book is a rectangle and yet also be seeing it very close to their face so that it's visual uh projection was was different and yet here there was different Behavior associated with the frame alone so what's going on there where even though they apparently can be reconstructed from each other what flips can we flip from thinking more euclidean and then in that situation um you know I'm a plane 30 000 feet above my city so there's nowhere I need to go but then in the projective setting we do have this kind of inbuilt epistemic drive to to be near so I think it's it's a hard coded so so the the idea is one of the things that it can be useful for is communication so using space as a way to communicate taking an example that we have very different architectures like the way that we treat information are not the same we don't have exactly the same connection neurons and everything but we still have like a common framework in which we can discuss for example in a visual visual information is something that is very immediate for us which is not obvious if you take the perspective the point of view if you go from the idea that it's rebuilt and it's not the environment you're living on it's not in it's not the exactly the space that we see there's something that we constructed like uh for uh like uh functional reasons and that is one of them could be that we can communicate very directly with it so I think it's hard coded that the state space needs to have this action and that all the agents share this action but they don't share the same architecture so it gives a common base for discussion like true actions for example change that of perspective so it gives like the so it's the chance of a question or clean projective we're not going in the direction uh in terms of research where we're trying to say that we can go from a community to projective for the same agent we so agents are hard-coded with the latent space which is a has some structure and then we try to export it for function for communication for multi-agent uh like uh Behavior so that they can collaborate this is okay so each agent is within its own projective setting and then the euclidean sets the stage and allows that multi-perspective the oxygen space which is the outside world is just here in this toy model because we it's a way to reference the configuration of the world but for a network this would be for example the configurations of its sensors so it's just here there's a there's this kind of ambiguity between space and space so outside space which is occasion and inside which is projective but the outside space is just a way to discuss uh configurations of something that is related to the environment so it can be like your your sensors the configuration of your sensors or something like this information coming from your sensors so you just give a state space from it and the geometry is really in the reconstructured world that is internal but the being able that you do networks that do this is somewhere that we're working on now and it's not that obvious because you need to there's a lot of algorithms the problem algorithmic problems that we try to solve and that we need to solve so in the story Model is really like the ambiguity comes only from the fact that it's a toy model and but what is space is inside it what is reconstructed is the thing that we perceive space from sensors as being a whole as being something that structures the information and so David wants to say something or well I mean yes indeed in the talk you are having too I was giving them also and it really felt if I remember well yeah indeed the toy models encoded a certain world model in an Acadian uh way uh and then using homogeneous coordinate and this kind of stuff to do the projective uh transformation uh we just transform the Acadian space into a projective space and you can go of course it's in you can invert it I mean there is an online air division but you can invert it anyway uh so you can go from one to the other but Gregor is right that it was a choice of modeling it was a bit motivated by my experiment my experience as a brain scientist you know there are reasons to believe that memory for instance Place sales and grid cells they then do their new hypothesis about their hyperbolic characters but they you know usually they are thought as encoding space in the nuclearian manner so since when we project uh you know future action we use sometimes and we have to use memory or even if we do a remembrance of the past and we project ourselves in the scenes in the past there is this access to memory systems that you know we used to think and probably this is still the case on code information in a Korean manner so it would make sense that then for the conscious access there will be some operation that would allow us to go from the Cleveland to project which you know was implemented that way at the same time this is way beyond my my abilities in math so regular will correct me there is a way of seeing projective geometry as being more General as an extension of a fine Space by adding point at Infinity so if now you think about the project space with a metric basically uh the projective space is an extension of a trillion space am I saying something completely false yeah I mean it's one way to sing it and also a lot of operations that use dialogues I would say exchange between projective and Acadian if you think about multi-view reconstruction of 3D equipment space for multiple shots in your camera you take several shots of a building from different perspective and then you can use a deterministic algorithm I mean approach using people are geometry to kind of infer under some fires that basically all lines that converge at Point at Infinity are actually parallel then you reconstruct you go from projective to euclidean so there is a you know a deep relationship between the two that is possible and probably functional yeah that's that's about it so and people are geometry is a is a subset of projective geometry no yeah so just to leave some context David has a better intuition on protective spaces and me this is like this so when he says stuff I I I I like I I not he really has a very uh intuitive like physical personal like visualization and most of the time he's right so yeah experiencing a projective geometry or what is it like to experience a projective geometry so one one problem that there's this with this model is the fact that you have so you need to force the point that Infinity to be um so the plane you take away from the projected space has to be behind and So you you're not really working with real Maps you when no you can always extend them that's okay but I mean the thing is if you want to see it in the way that we see it s which is we have something in front of us I find it a bit restrictive to say that you cannot see what's behind you I mean you never see it right I mean you can imagine it which is not the same thing because you change your frame so I can't answer really this question I'm just pointing out the fact that for me it's very perturbating to think that I have a plane that corresponds to uh this continuity uh I mean with respect to I mean I have a plane that destroys completely the way that I can think about movement behind me which is that if I had to imagine I had the whole space in that is around me and not just in front of me things would be very very weird in terms of Transformations but it's not something that is not possible possible I mean it's uh this is what happens when you take psychiatric you basically allows the projective transformation yeah like maybe you have all these degrees of freedom but in the perception like in some sorry motor processes this is calibrated this is restricted so we have to choose certain subset of the group basically it's just it's only certain project transformation that will be used in practice but now if you go into mystical experience or you take drugs they are going to mess precisely that's the hypothesis because with those parameters and now you have the full flight 15 degrees of freedom transformation and you see things that are very weird like that looks like mandalas or things like that so that's something you can get when you when you leave too much freedom to the projective action basically it's having yeah what if what we're seeing quote in front of us is behind us or something like that what do you mean what was the what was the issue with not being able to see behind us the issue is so if you want to see the project space as a 3D space as a way that we skip now need to take away your plane and basically projective transformation so that is sort of playing at Infinity on which you can I mean it doesn't exist in the projected space it exists only in the way you define the navigation chart like a 3D like our R3 chart uh with the plane that is taken away and and so this plane that is behind you if you apply a projective transformation like in front of you everything will seem fine you're You're Expecting like it's things that you experience but then once what happened is that it can send things at the infinity back behind you and so this is something that is a bit I mean for me I'm very on on I mean it's if I so you can write it explicitly there's no problem mathematically but if it's really the way that we experience it it's extremely weird to think about so I prefer not thinking about it so this is the only thing I and I don't think drugs so I'll leave it to other people I mean it's really a good it's true that taking drugs in this context makes complete like you can you could model it in the projective framework and not in the euclidean framework because you have a huge amount of protective Transformations that make sense more than the one that are related to actions in there so in in the presentation I give there was a way to relay attack like a cleaning friends to protect the frames under certain actions and in fact you have a huge much bigger space of projective Transformations than the one that are restricted through by uh the actions of the agents so the action of the agent inside of the real space so the occlusion space induced projective transformation in its internal world but there are much more projective transformation than these actions and so you can imagine in this setting having like very confusing States uh of your mind in a way it's really a good friend like this which you cannot in the application case I wonder if if a creature with or a robot with 360 degree cameras would it not be so perplexed as it wouldn't necessarily have a visual before or a visual in front and behind and the thing is how do you glue them together they still have to pick a point you still have to pick a point of convergence right I mean you need to you need to find a way to do multi-modal integration like if you if you want to have a camera that's 360 uh you have if you have the space to represent it so maybe you can represent it as a sphere and in this case the sense the sensors the sensors are different so it's it's not necessarily the same the same robot than the models that we are considering but there might be also another homogeneous space so a g space which is a space acting like which corresponds to the rotations of the uh of uh of uh like if you say that it can look 360 and then you can rotate and the space of observation stays the same so you can always imagine acting on it and so you can imagine like having another g-space structure and the good thing about it is just with respect to the perspective if you have an agent that can take a different perspective on this environment which depends on its sensors and its actions and the kind of data you have then you can Define G space and you can do exactly the same setting everything like it's it's made in the morals General more General context so you can apply it and so it's not limited only to the projective case I like we come from the PCM projective Consciousness model but uh I mean the the framework now are at least the way to replace this framework inside of uh optimal control is uh well adapted for any change of perspective and nothing and not just awesome well Where Are You Gonna go to next where will the epistemic drive take you carrying forward so that there are two projects now so the first one is to be able to be like have a maybe that we can talk but but is to be able to uh to get uh so to implement it so that it can be used for uh monitoring behaviors or to be able to predict behaviors or to analyze it so if you have a model of behaviors of Agents like maladaptive behaviors think that we already published but in more limited context now to make it inside of a computational framework so that we can use it to analyze like experiments with humans so this is the first project which is ongoing with several students so on the page that I put on my on the first slide there's all the work that we're doing the people who are working with us and we're very grateful for them working with us and all the ongoing work in this direction and there's the second one which is more like say machine learning and so for me it's very important you have effective algorithms things I mean like things that you can really use in practice things that you can really Implement and implementing when you have group structures there are many many problems that appear how to do sarcastic optimal control when the latent space is a homogeneous space it's I think a completely open question when I start you have answers in this direction so I hope hopefully the the work will go out come out soon let's say uh January or something like this but it's there's a huge work in this direction cool anything else you want to add thank you very much for the invitation a lot to think about Slash mentally rotate slash project I mean after all as you even alluded to with communication yeah this is a perspective taking question Technologies adding some new flavors and methods for example asynchronous communication but it's synchronous for the agent when they perceive it but all these different modalities of communication and it's quite interesting to think about and just really cool that you and colleagues are pursuing that from empirical data analytic and from theoretical mathematical approaches I mean on this side for communication I think and so it's something I I find extremely interesting like more generally if some people want to some people want to work on the project please tell us and I will be very happy and if you and even if you have money please tell us and who are very happy here so I agree great great we were very open you can always send us a mail we'll be very happy to discuss about this week okay well thank you again to to um all till next time see ya bye bye